

Gatuline® Age Defense NP

Beauty armor



People make our name



Proactive shield



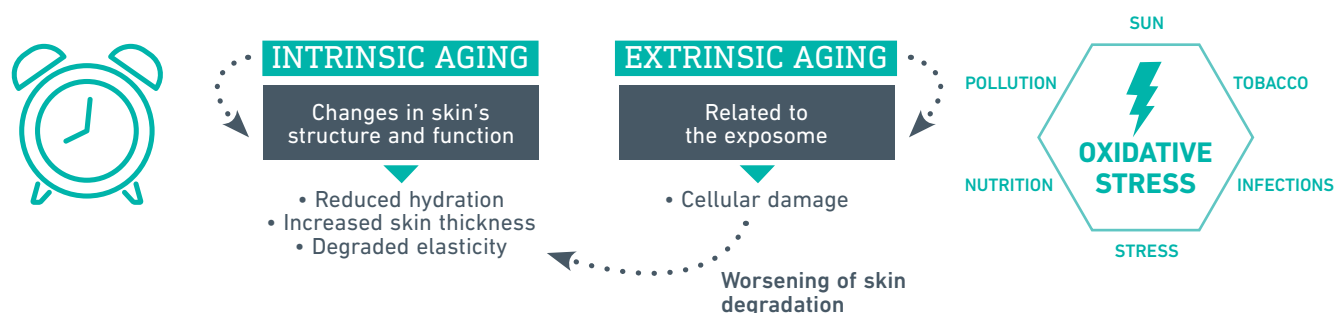
The need for greater sustainable development has changed the industry's approach to the supply chain, and sourcing and production processes of raw materials are shifting to align with the new paradigm.

Embracing this shift, Gatuline® Age Defense NP emerged from an original idea to valorize a by-product of local agriculture and handicraft.

Based on wasted solid residue obtained after the extraction of walnut oil, Gattefossé has created a unique active ingredient protecting the skin from external and internal aging.

Skin aging explained

Aging is a physiological process, genetically programmed and aggravated by external events. Several modifications of the cutaneous tissues occur over time and have two origins:

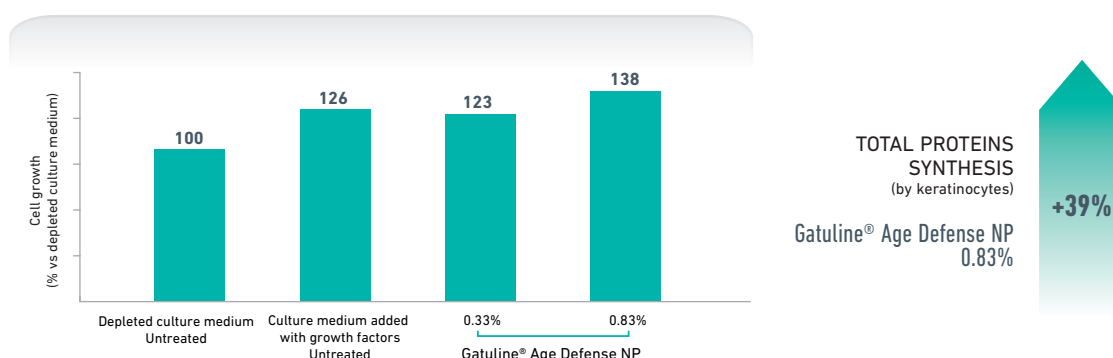


When the skin ages and is damaged by external aggressions, a natural anti-oxidant pool counteracts the effect of the reactive oxygen species. But this system can become saturated and then fails to neutralize the free radicals in excess. A double protection against time and environmental assaults appears to be a winning strategy to preserve skin youth.

Protection against time

Gatuline® Age Defense NP features biostimulating capacities. By boosting cells' metabolism, it counteracts the slowdown of cellular functions induced with age.

It has been tested on human keratinocytes, compared to a culture medium added with growth factors in order to measure the cells ability to growth and synthesize proteins.

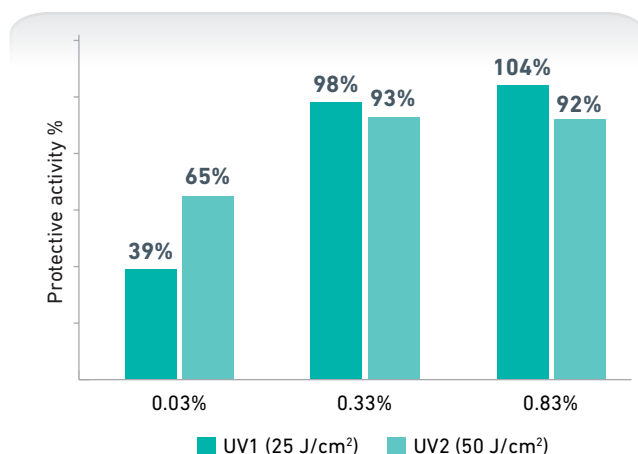


Gatuline® Age Defense NP **stimulates keratinocytes proliferation and boosts the amount of proteins synthesized.** The neosynthesis of proteins has also been measured on fibroblasts (substantiated).

A natural antioxidant

Protection against oxidative stress

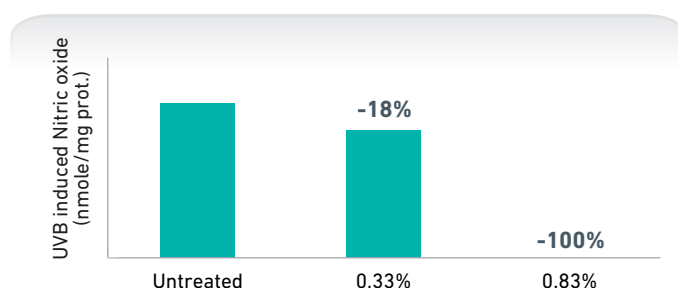
The protective activity of Gatuline® Age Defense NP has been assessed on keratinocytes, by measuring the intracellular oxidation rate following UVB exposure.



Gatuline® Age Defense NP **protects the cells against oxidative stress**, with a dose dependent effect.

The release of pro-inflammatory mediators, such as nitric oxide is a sign of aggressions and a direct consequence of the oxidative stress.

Its production has been measured on normal human keratinocytes after UVB exposure.

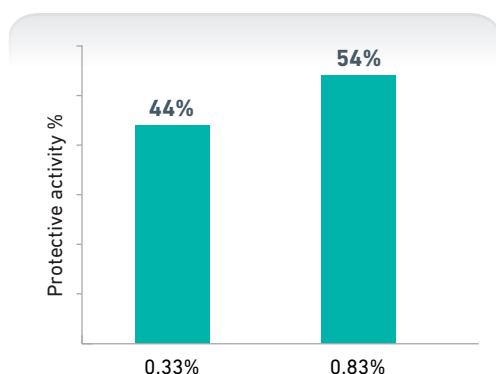


Gatuline® Age Defense NP completely inhibits nitric oxide production, showing clear **protective capacity against the consequences of oxidative stress**.

Prevention of UV-induced apoptosis

Apoptosis is a process of programmed cell death. It normally occurs during aging but it is also a defense mechanism when cells are damaged by noxious agents.

The protective activity against UVB-induced apoptosis was evaluated on keratinocytes.



Gatuline® Age Defense NP **protects the skin from UVB-induced apoptosis**.

Boosted protection for the hair and the skin

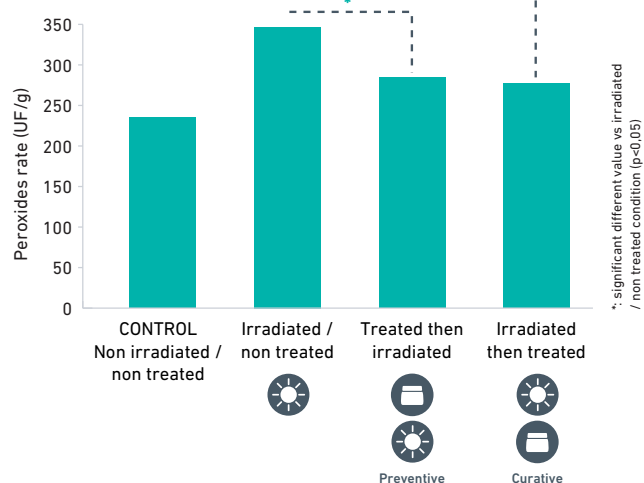
The scavenging potential of Gatuline® Age Defense NP has been demonstrated *in vivo* on skin and *ex vivo* on hair locks through the measurement of the peroxide rate.

ANTIOXIDANT PROTECTION



ONTO THE HAIR ⁽¹⁾

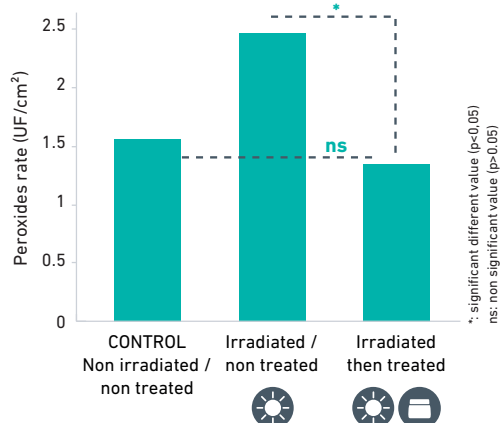
Protects against hair peroxide formation



Gatuline® Age Defense NP **protects the hair fiber against the deleterious effects of UV** both preventively and curatively.

ONTO THE SKIN ⁽²⁾

Protects against skin peroxide formation



Gatuline® Age Defense NP facilitates the return to a basal level of cutaneous peroxide, proving that it **helps to prevent oxidative stress, one of the major causes of skin aging.**

⁽¹⁾ Clinical evaluation on 10 hair locks soaked in an aqueous solution containing 0.5% Gatuline® Age Defense NP before and after UVA/UVB irradiation

⁽²⁾ Clinical evaluation on 10 panelists, 18 to 40 years old, application of a gel containing 6% Gatuline® Age Defense NP on the back after UVA/UVB exposure



Gatuline® Age Defense NP

Rich in scavenging and energizing molecules, Gatuline® Age Defense NP protects the natural cellular antioxidant pool. The skin is protected from harsh external factors—e.g. UV and pollution—and able to resist the effects of time.

Gatuline® Age Defense NP is a natural origin active ingredient extracted from walnut seed. Walnuts are harvested in the Rhône-Alpes region (less than 100 km from Gattefossé manufacturing site). A long-time partnership with a local farmer enables a perfect traceability of the raw material from the field to the lab.

A true youth treasure, this superfood ingredient possesses a rich phytochemical composition. Our extract is particularly rich in phytic acid and polyphenols, two powerful chelating agents able to prevent oxidative stress.

Claims

- Provides complete protection from oxidative stress
- Boosts cell metabolism
- Shields from the effects of pollution
- Fights extrinsic and intrinsic aging
- Protects the hair and skin from UV and environmental damage

Applications

- Anti-aging and protective treatments
- Sun treatments
- Anti-pollution and anti-stress products
- Protective hair treatments

Use level

0.5-6%

INCI name: Water (and) Juglans Regia (Walnut) Seed Extract

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2015

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